

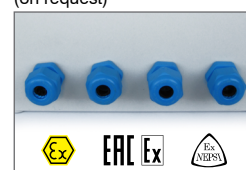
W200BOX

WEIGHT INDICATOR INTO BOX - WEIGHING AND BATCHING

LAUMAS®



ATEX/EAC EX version
(on request)



PROGRAM

CODE

BASE	W200BOX-B
LOAD	W200BOX-C
UNLOAD	W200BOX-S
3 PRODUCTS	W200BOX-3
* 6 PRODUCTS	W200BOX-6
* 14 PRODUCTS	W200BOX-14
Multiprogram	W200BOX-MU

* External 8-relay modules included.

FIELDBUSES

MODBUS RTU
MODBUS/TCP

CANopen

PROFIBUS

DeviceNet

EtherNet/IP

ETHERNET
TCP/IP

PI CERTIFIED
PROFIBUS • PROFINET

Rev. 0.0

CERTIFICATIONS

	OIML R76:2006, class III, 3x10000 divisions, 0.2 μ V/VSI / OIML R61 - WELMEC Guide 8.8:2011 (MID)
	UL Recognized component - Complies with United States and Canada regulations
	Complies with the Eurasian Customs Union regulations
	Equivalent of the CE marking for the United Kingdom
	NMI Trade Approved - Complies with Australian market regulations for legal for trade use
	Complies with New Zealand regulations for legal for trade use
	Complies with the Brazilian regulations for legal for trade use
	NTEP - n_{max} 10000 - Class III/IIIL - Complies with United States regulations for legal for trade use
	Complies with Chinese market regulations for legal for trade use

CERTIFICATIONS ON REQUEST

	Conformity assessment (initial verification) in combination with Laumas weighing module
	ATEX (zone 2-22) → The external relay modules must be protected.
	Complies with the Eurasian Customs Union standards for use in potentially explosive atmospheres
	Complies with Chinese market regulations for use in potentially explosive atmospheres
	Complies with the regulations of the Russian Federation for legal for trade use

DESCRIPTION

- Weight indicator in IP67 polycarbonate box with 4+2 M16x1.5 cable glands-plugs, suitable for wall mounting.
- Dimensions: 170x140x95 mm (4 fixing holes Ø 4 mm; centre distance: 152x122 mm).
- 6-digit semi-alphanumeric red LED display (14 mm height).
- 8 signalling LED.
- 5-key keyboard.
- Real-time clock/calendar with buffer battery.
- The instrument can be configured and managed using the free "Instrument Manager" PC software, which you can download from www.laumas.com.

INPUTS/OUTPUTS AND COMMUNICATION

- RS485/RS232 serial ports for communication via protocols ModBus RTU, ASCII Laumas or continuous one way transmission.
- 5 relay outputs controlled by the setpoint values or via protocols (4 outputs if analog output is present).
- 3 optoisolated PNP digital inputs: status reading via serial communication protocols (2 inputs if analog output is present).
- 1 load cell dedicated input.
- Current or voltage 16 bit optoisolated analog output (option on request).

MAIN FUNCTIONS

- Connections to:
 - PLC via analog output (on request);
 - PC/PLC via RS485/RS232 (up to 99 instruments with line repeaters, up to 32 without line repeaters);
 - remote display and printer via RS485/RS232;
 - up to 8 load cells in parallel by junction box;
 - intelligent junction box or other multichannel instruments: allow the use of advanced functions as digital equalization, load distribution analysis and automatic diagnostics.
- TCP/IP WEB APP: integrated software in combination with the Ethernet TCP/IP option for remote supervision, management and control of the instrument.
- Digital filter to reduce the effects of weight oscillation.
- Theoretical calibration (via keyboard) and real calibration (with sample weights and the possibility of weight linearization up to 8 points).
- Tare weight zero setting.
- Automatic zero setting at power-on.
- Gross weight zero tracking.
- Semi-automatic tare (net/gross weight) and preset tare.
- Semi-automatic zero.
- Displaying of the maximum weight value reached (peak).
- Direct connection between RS485 and RS232 without converter.
- Weight value printing with date and time via keyboard or external contact.
- Printing (header) can be customized using the free "JollyPrint" PC software, available at www.laumas.com.
- Labeling machine management (except 3/6/14 PRODUCTS program).

Approved versions for legal for trade use

- System parameters management protected by qualified access via software (password), hardware or fieldbus.
- Weight subdivisions displaying (1/10 e).
- Three operation mode: single interval or multiple range or multi-interval.
- Net weight zero tracking.
- Calibration.
- Alibi memory (option on request).
- The following values can be printed via keyboard or external contact: gross weight, net weight, tare, preset tare, date, time, ID code (alibi memory).

BASE PROGRAM

- Hysteresis and setpoint value setting.
- The indicator can be used as a remote display with setpoints.
- 12 groups selection by 5 setpoint via external selector switch or contact (option on request).

BATCHING PROGRAM

- 99 settable formulas.
- Batching resume after a blackout.
- Automatic fall calculation.
- Tolerance error control.
- Precision batching through slow function.
- Precision batching through tapping function.
- Consumption storage.
- Printing of batching data.
- Alarm contact management.
- Selection of the first 12 formulas via external selector switch or contact (option on request).
- Batching start via external contact or keyboard.
- Manual batching with remote displays connected in parallel to the instrument.

Only for:

LOAD and 3/6/14 PRODUCTS programs

- Autotare at batching start.
- Setting a quantity to be batched greater than the scale capacity.

UNLOAD program

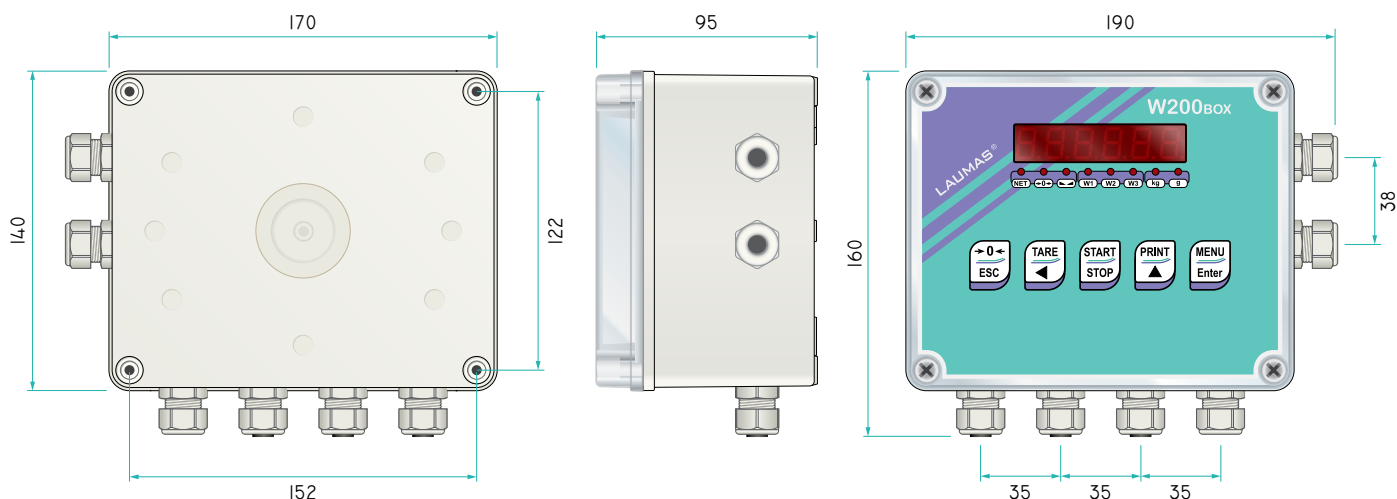
- Automatic loading of the product into the weighed structure.
- Management of batching with big bags.

3/6/14 PRODUCTS program

- Formulas programming in fixed or variable steps.
- Formulas setting in percentage.
- Intermediate unloadings during the batching.
- Partial unloadings at cycle end.

MULTIPROGRAM

- The Multiprogram instruments do not have any selected program but can be set by the installer with different operating modes: BASE, LOAD, UNLOAD, 3 PRODUCTS, 6 PRODUCTS, 14 PRODUCTS.



TECHNICAL FEATURES

Power supply and consumption	12÷24 VDC ±10%; 5 W
Number of load cells • Load cells supply	up to 8 (350 Ω) - 4/6 wires • 5 VDC/240 mA
Linearity • Analog output linearity	<0.01% full scale • <0.01% full scale
Thermal drift • Analog output thermal drift	<0.0005% full scale/°C • <0.003% full scale/°C
A/D Converter	24 bit (16000000 points) - 4.8 kHz
Divisions (with measurement range ±10 mV and sensitivity 2 mV/V)	±999999 • 0.01 μV/d
Measurement range	±39 mV
Usable load cells sensitivity	±7 mV/V
Conversions per second	300
Display range	±999999
Decimals • Display increments	0÷4 • x1 x2 x5 x10 x20 x50 x100
Digital filter • Readings per second	10 levels • 5÷300
Relay outputs	5/4 - max 115 VAC/150 mA
Optoisolated digital inputs	3/2 - 5÷24 VDC PNP
Serial ports	RS485, RS232
Baud rate	1200, 2400, 4800, 9600, 19200, 38400, 115200 (bit/s)
Optoisolated analog output (option on request)	16 bit = 65535 divisions. 0÷20 mA; 4÷20 mA (up to 300 Ω) 0÷10 V; 0÷5 V; ±10 V; ±5 V (min 10 kΩ)
Humidity (condensate free)	85%
Storage temperature	-30 °C +80 °C
Working temperature	-20 °C +60 °C

	Relay outputs	5/4 - max 30 VAC, 60 VDC/150 mA
	Working temperature	-20 °C +50 °C
	Equipment to be powered by 12-24 VDC LPS or Class 2 power source	

METROLOGICAL SPECIFICATIONS OF TYPE-APPROVED INSTRUMENTS

	OIML	NTEP	INMETRO
Applied standards by region	EU: 2014/31/UE; OIML R76:2006; EN45501:2015 Russian Federation: GOST OIML R76-1-2011 United Kingdom: Non-automatic Weighing Instrument Regulations 2016 Australia: National Measurement Regulations 1999 New Zealand: Weights and Measures Regulations 1999 China: Law on Metrology of the People's Republic of China	USA: NIST HANDBOOK 44, 2020; NCWM PUB 14, 2021	Brazil: Portaria Inmetro N°157/2022
Operation mode	single interval, multi-interval, multiple range	single interval, multi-interval, multiple range	single interval, multi-interval, multiple range
Accuracy class	III or IIIL	III or IIIL	III
Maximum number of scale verification divisions	10000 (class III); 1000 (class IIIL)	10000 (class III/IIIL)	10000 (class III)
Minimum input signal for scale verification division	0.2 μV/VSI		0.2 μV/VSI
Working temperature	-10 °C +40 °C	-10 °C +40 °C (+14 °F +104 °F)	-10 °C +40 °C

OPTIONS ON REQUEST AND COMPATIBILITY WITH BATCHING PROGRAMS

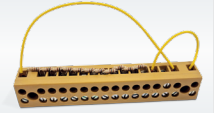
INTERFACES AND FIELD BUSES

 ANALOG OUTPUT	* Optoisolated 16 bit analog output. ➔ One input and one output not available.
 RS485	* Additional RS485 port. ➔ One input and one output not available. ➔ Not compatible with E/EC option.
 CANopen	* CANopen protocol.
 DeviceNet	* DeviceNet protocol.
 PROFIBUS	* Profibus DP protocol.
 Ethernet/IP	* Ethernet/IP protocol - Ethernet port. ➔ Internal crimp wiring.
 ETHERNET TCP/IP	* Ethernet TCP/IP protocol - Ethernet port. - Integrated software for remote supervision, management and control of the instrument. ➔ Internal crimp wiring.
 MODBUS/TCP	* Modbus/TCP protocol - Ethernet port. ➔ Internal crimp wiring.
 PIV PROFIBUS • PROFINET	* Profinet IO protocol - Ethernet port. ➔ Internal crimp wiring.
 0-10	Weight reading from 0-10 VDC input (15 kΩ).
 4-20	Weight reading from 4-20 mA input (120 Ω).

* Select one option among those marked with an asterisk.

OPTIONS ON REQUEST AND COMPATIBILITY WITH BATCHING PROGRAMS

EXPANSIONS

	★ EC Base: 12 groups selection by 5 setpoint via external selector switch. Load, Unload, 3/6/14 Products: selection of the first 12 formulas via external selector switch.				
	★ E Base: 12 groups selection by 5 setpoint via external contact. Load, Unload, 3/6/14 Products: selection of the first 12 formulas via external contact.				
	Simultaneous use of E/EC option with the analog output.				
	External 5-relay module to increase the capacity of SPDT contacts to 115 VAC/2 A. → <i>Not compatible with models 6/14 PRODUCTS.</i>				
	External 8-relay module to manage from 1 to 6 products; 8 relays up to max 115 VAC/2 A. Module included with models 6/14 PRODUCTS. <table border="0"> <tr> <td>RELE6PROD24V</td><td>12 ÷ 24 VDC</td></tr> <tr> <td>RELE6PROD230V</td><td>115/230 VAC</td></tr> </table>	RELE6PROD24V	12 ÷ 24 VDC	RELE6PROD230V	115/230 VAC
RELE6PROD24V	12 ÷ 24 VDC				
RELE6PROD230V	115/230 VAC				
	RELE14PROD External 8-relay module to manage from 7 to 14 products to be added to RELE6PROD module; 8 relays up to max 115 VAC/2 A. Module included with model 14 PRODUCTS.				

★ Select one option among those marked with an asterisk.

OPTIONS ON REQUEST AND COMPATIBILITY WITH BATCHING PROGRAMS

APPLICATIONS - SOFTWARE

	Alibi memory.
	DATAPC Data transfer from the instrument to the PC, via RS232 (directly) or RS485 (by converter) serial port. These data (weighed values, batchings, alarms) can be imported and processed on the PC using the PROG-DB software included. We suggest to use this option when the indicator is always connected to the PC.